Pfizer-BioNTech COVID-19 Vaccine

EUA 27034

Response to CBER Information Request Received on 04 December 2020 Regarding Dataset Variable Derivations in Emergency Use Authorization Submission

December 2020

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1. INTRODUCTION

Reference is made to Request for Emergency Use Authorization (EUA 27034) submitted 20 November 2020 by Pfizer on behalf of BioNTech. Further reference is made to CBER's 04 December 2020 Information Request communicated by email from (b) (6) (b) (6) (CBER), to Elisa Harkins (Pfizer, Inc.).

CBER comments are presented in ***bold italics*** and Pfizer-BioNTech's responses follow in plain text.

2. CBER COMMENT

Derived variables BE1DT, BE1DT2, BE2DT and BE2DT2 were used to derive ADSL.V01DT and ADSL.V02DT for safety analyses. However, it is unclear how these four variables were derived. Please explain the programming logic and provide the code used for derivation. If possible, please submit a data set with these four variables, or clarify which submitted data set includes them, if they have already been submitted.

<u>Response</u>

The four variables, BE1DT, BE1DT2, BE2DT and BE2DT2, are temporarily derived variables in ADSL. They were used to derive ADSL.V01DT and ADSL.V02DT for safety analyses. Please find the programming logic and derivation code below. All source/Input datasets were submitted in the eSUB SDTM package. Please note LIBNAME "dataprot" needs to be defined based on FDA's SDTM source data location.

<table>
<tr><th>Variable</th><th>Label</th><th>Input SDTM Datasets</th><th>Derivation Logic/Code</th></tr>
<tr><td>BE1DT</td><td>1-Month post dose 2 blood sample drawn date</td><td>CO/ SUPPDS</td><td>Programming Logic:
1-Month post dose 2 blood sample drawn date from CO.CODTC where ROMAIN='IS' and COREF="Sample Collected" and COVAL="Y";

Note: 'V7_MONTH1_S' is the corresponding visit name for phase 1 subjects except for the 100 ug dose cohort. 'V3_MONTH1_POSTVAX2_L' is the corresponding visit name for phase 2/3 subjects.
For phase 1 100 ug dose cohort (cohort=1.16), the 2nd dose 10 ug was taken about 3 monthes after dose 1 and 'V7_MONTH1_S_R' is the corresponding 1-month after the 2nd dose visit name.

Programming Code:
<pre>proc sql;
  create table isva as
  select a.*, case when index(b.qval,"100mcg")>0 then 1.16 end as cohortn
  from dataprot.co(where=(RDOMAIN="IS" and COREF="Sample Collected" and
COVAL="Y")) a
  left join dataprot.suppds(where=(qnam="DSRANGRP")) b
  on a.usubjid=b.usubjid;
quit;

data be1dt;
  set isva;
  where (visit in ('V3_MONTH1_POSTVAX2_L', 'V7_MONTH1_S') and cohortn ne
1.16) or
 (visit in ('V7_MONTH1_S_R') and cohortn=1.16);/*For phase 1 100 ug cohort*/
  be1dt=input(codtc, yymmdd10.);
run;</pre></td></tr>
<tr><td>BE2DT</td><td>6-Month post dose 2 blood sample drawn date</td><td>CO/ SUPPDS</td><td>Programming Logic:
6-Month post dose 2 blood sample drawn date from CO.CODTC where ROMAIN='IS' and COREF="Sample Collected" and COVAL="Y";

Note: 'V8_MONTH6_S' is the corresponding visit name for phase 1 subjects except for the 100 ug dose cohort. 'V4_MONTH6_L' is the corresponding visit name for phase 2/3 subjects.
For phase 1 100 ug dose cohort (cohort=1.16), the 2nd dose 10 ug was taken about 3 monthes after dose 1 and 'V8_MONTH6_S_R' is the corresponding 6-month after the 2nd dose visit name.

Programming Code:</td></tr>
</table>

<table>
<tr><td></td><td></td><td></td><td>Note: data set isva is obtained from above steps.
<pre>data be2dt;
  set isva;
  where (visit in ('V8_MONTH6_S', 'V4_MONTH6_L') and cohortn ne 1.16) or
(visit in ('V8_MONTH6_S_R') and cohortn=1.16); /*For phase 1 100 ug cohort*/
  be2dt=input(codtc, yymmdd10.);
run;</pre></td></tr>
<tr><td>BE1DT2</td><td>1-Month post dose 2 follow up visit date</td><td>SV/ SUPPDS</td><td>Programming Logic:
1-Month post dose 2 visit date from SV.SVSTDTC

Programming Code:
<pre>proc sql;
  create table sv as
  select a.*, case when index(b.qval,"100mcg")>0 then 1.16 end as cohortn
  from dataprot.sv a
  left join dataprot.suppds(where=(qnam="DSRANGRP")) b
  on a.usubjid=b.usubjid;
quit;

data be1dt2;
set sv;
where (visit in ('V3_MONTH1_POSTVAX2_L', 'V7_MONTH1_S') and cohortn ne 1.16)
or
 (visit in ('V7_MONTH1_S_R') and cohortn=1.16); /*For phase 1 100 ug
cohort*/
  be1dt2=input(svstdtc, yymmdd10.);
run;</pre></td></tr>
<tr><td>BE2DT2</td><td>6-Month post dose 2 follow up visit date</td><td>SV/ SUPPDS</td><td>Programming Logic:
6-Month post dose 2 visit date from SV.SVSTDTC

Programming Code:
Note: data set sv is obtained from above steps.
<pre>data be2dt2;
  set sv;
  where (visit in ('V8_MONTH6_S', 'V4_MONTH6_L') and cohortn ne 1.16) or
(visit in ('V8_MONTH6_S_R') and cohortn=1.16); /*For phase 1 100 ug cohort*/
  be2dt2=input(svstdtc, yymmdd10.);
run;</pre></td></tr>
</table>

V01DT	Date of Visit at 1-Month after Vax2 for safety analysis	CO/ SV/ EX/ DM/ SUPPDS	**Programming Logic:**
			For Phase 1:
			V01DT is the blood sample drawn date from visit 7;
			If visit 7 blood sample drawn date is not available from CO dataset, then use the date of visit 7 from SV dataset.
			Else if date of visit 7 from SV dataset is not available, then use date of dose 2 + 35 days;
			Else if date of dose 2 is not available, then use date of dose 1 + 58 days.
			
			For Phase 2/3:
			V01DT is the blood sample drawn date from visit 3;
			If visit 3 blood sample drawn date is not available from CO dataset, then use the date of visit 3 from SV dataset.
			Else if date of visit 3 from SV dataset is not available, then use date of dose 2 + 35 days;
			Else if date of dose 2 is not available, then use date of dose 1 + 58 days.
			
			Note: There are two subjects took unplanned dose 3 and for these two subjects, dose 3 date was used to derive V01DT (V01DT= date of dose 3 + 35 days).
			
			`[VAX101DT = dose 1 date from EX`
			`VAX102DT = dose 2 date from EX`
			`VAX103DT = dose 3 date from EX: unplanned dose 3]`
			
			Programming Code:
			Note: data sets be1dt/be2dt/be1dt2/be2dt2 are obtained from above steps.
			
			`proc sort data=dataprot.ex out=ex nodupkey;`
			`  by usubjid extptref exstdtc;`
			`  where usubjid ne "" and exstdtc ne "";`
			`run;`
			
			`proc sort data=ex;`
			`  by usubjid exstdtc exendtc visitnum;`
			`run;`
			
			`data ex;`
			`  set ex;`
			`  by usubjid exstdtc exendtc visitnum;`
			`  retain extptref1n;`
			`  if first.usubjid then`
			`    extptref1n=1;`
			`  else`
			`    extptref1n+1;`

			`` format exdate date9.;
			  exdate=input(scan(exstdtc, 1, "T"), yymmdd10.);
			run;
			
			proc transpose data=ex out=t_ex prefix=vax10 suffix=dt;
			  by usubjid;
			  id extptref1n;
			  var exdate;
			run;
			
			data adsl;
			  merge dataprot.dm(in=a) t_ex;
			  by usubjid;
			  if a;
			run;
			
			data adsl;
			  merge adsl(in=a) be1dt(keep=usubjid be1dt) be2dt(keep=usubjid be2dt)
			        be1dt2(keep=usubjid be1dt2) be2dt2(keep=usubjid be2dt2);
			  by usubjid;
			  if a;
			
			  if not missing(VAX103DT) then
			    V01DT=VAX103DT+35;/*Only for the two subjects took unplanned dose 3*/
			  else if not missing(be1dt) then
			    V01DT=be1dt;
			  else if not missing(be1dt2) then
			    V01DT=be1dt2;
			  else if not missing(VAX102DT) then
			    V01DT=VAX102DT+35;
			  else if not missing(VAX101DT) then
			    V01DT=VAX101DT+58;
			
			  if not missing(VAX103DT) then
			    V02DT=VAX103DT+168;/*Only for the two subjects took unplanned dose 3*/
			  else if not missing(be2dt) then
			    V02DT=be2dt;
			  else if not missing(be2dt2) then
			    V02DT=be2dt2;
			  else if not missing(VAX102DT) then
			    V02DT=VAX102DT+168;
			  else if not missing(VAX101DT) then```

<table>
<tr><td></td><td></td><td></td><td><pre>   V02DT=VAX101DT+191;

  format V01DT V02DT date9.;
  label V01DT="Date of Visit at 1M after Vax2"
        V02DT="Date of Visit at 6M after Vax2";
  drop BE1DT BE2DT BE1DT2 BE2DT2;
run;</pre></td></tr>
<tr><td>V02DT</td><td>Date of Visit at 6-Month after Vax2 for safety analysis</td><td>CO/SV/EX/ SUPPDS</td><td>Programming Logic:
For Phase 1:
V02DT is the blood sample drawn date from visit 8.
If visit 8 blood sample drawn date is not available from CO dataset, then use the date of visit 8 from SV dataset.
Else if date of visit 8 from SV dataset is not available, then use date of dose 2 + 168 days
Else if date of dose 2 is not available, then use date of dose 1 + 191 days

For Phase 2/3:
V02DT is the blood sample drawn date from visit 4. /*No subject with this date at the date of cutoff for EUA*/
If visit 4 blood sample collection date is not available from CO dataset, then use the date of visit 4 from SV dataset. /*No subject with this date at the date of cutoff for EUA*/
Else if date of visit 4 from SV dataset is not available, then use date of dose 2 + 168 days
Else if date of dose 2 is not available, then use date of dose 1 + 191 days

Note: There are two subjects took unplanned dose 3 and for these two subjects, dose 3 date was used to derive V02DT (V02DT= date of dose 3 + 168 days).
<pre>[VAX101DT = dose 1 date from EX
 VAX102DT = dose 2 date from EX
 VAX103DT = dose 3 date from EX: unplanned dose 3]</pre>Programming Code:
Note: refer to the complete code for V01DT above.
<pre>if not missing(VAX103DT) then
  V02DT=VAX103DT+168; /*Only for the two subjects took unplanned dose 3*/
else if not missing(be2dt) then
  V02DT=be2dt;
else if not missing(be2dt2) then
  V02DT=be2dt2;
else if not missing(VAX102DT) then
  V02DT=VAX102DT+168;</pre></td></tr>
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			`else if not missing(VAX101DT) then` `  V02DT=VAX101DT+191;`

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data be1dt;
  set isva;
  where (visit in ('V3_MONTH1_POSTVAX2_L', 'V7_MONTH1_S') and cohortn ne
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 (visit in ('V7_MONTH1_S_R') and cohortn=1.16);/*For phase 1 100 ug cohort*/
  be1dt=input(codtc, yymmdd10.);
run;</pre></td></tr>
<tr><td>BE2DT</td><td>6-Month post dose 2 blood sample drawn date</td><td>CO/ SUPPDS</td><td>Programming Logic:
6-Month post dose 2 blood sample drawn date from CO.CODTC where ROMAIN='IS' and COREF="Sample Collected" and COVAL="Y";

Note: 'V8_MONTH6_S' is the corresponding visit name for phase 1 subjects except for the 100 ug dose cohort. 'V4_MONTH6_L' is the corresponding visit name for phase 2/3 subjects.
For phase 1 100 ug dose cohort (cohort=1.16), the 2nd dose 10 ug was taken about 3 monthes after dose 1 and 'V8_MONTH6_S_R' is the corresponding 6-month after the 2nd dose visit name.

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Note: data set isva is obtained from above steps.</td></tr>
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Document Approval Record

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Signed By:	**Date(GMT)**	**Signing Capacity**
(b) (6) (b) (4)	07-Dec-2020 21:04:32	Business Line Approver